UNIVERSITY OF WOLVERHAMPTON

Gamified DSA Application

HERALD COLLEGE KATHMANDU

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Introduction

- Web-based interactive DSA platform with an intuitive, beginner-friendly design (no prior coding needed)
- Puzzle-driven learning approach replacing traditional lectures with engaging challenges
- Visual and hands-on exploration of arrays, stacks, trees, and graphs through direct interaction
- Gamified progression system with levels, points, badges, and real-world problem-solving simulations

Aims

1. Make DSA learning accessible and beginner-friendly
2. Teach concepts through interactive, puzzle-based gameplay

Objectives

1. Build a responsive platform with progress tracking and gamification
2. Create guided modules with hints, feedback, and assessment.

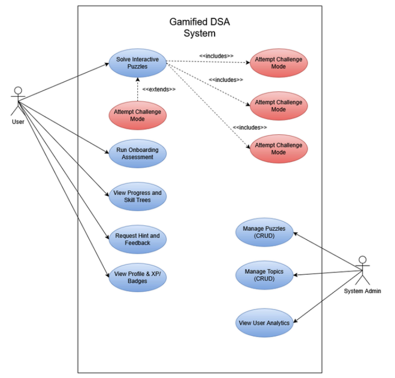
Literature Review

Study	Focus Area	Key Finding	Limitation	Relevance
Mtaho & Mselle (2024)	DSA Difficulty	High cognitive load due to abstract concepts	No concrete solution	Justifies need for visualization & interaction
Duran et al. (2025)	Active Learning	Improves engagement & understanding	Needs structured design	Supports interactive, hands-on learning
Camacho-Sánchez et al. (2022)	Gamification	Boosts motivation & performance	Not DSA-specific	Supports rewards & feedback system
Manzano-León et al. (2021)	Gamification Review	Improves engagement & outcomes	Mostly basic implementations	Need for deeper gamified design
Kratochvíl et al. (2023)	Novelty Effect	Engagement drops over time	Short-term focus	Need for evolving challenges

Academic Questions

How does gamification impact students' conceptual understanding and retention of abstract DSA topics?

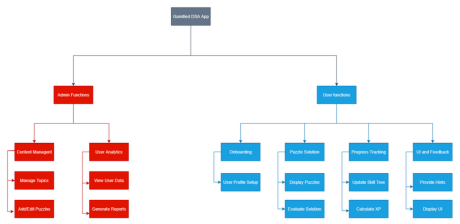
Use Case Diagram



Developed Product

- Puzzle-based, interactive learning with visual data structure manipulation
- Skill tree with progressive unlocking of DSA concepts
- Real-time feedback, guided hints, and gamified rewards (XP, badges, levels)
- User profile with progress tracking and responsive UI

Functional Decomposition Diagram



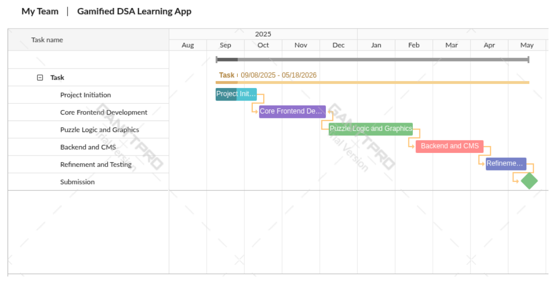
Testing

Manual Black-Box Functional Testing

- Focused on user inputs → outputs
- Ensured all major app features work as intended
- Identified bugs & UX issues for improvement

TC_005	**Skill Tree**	Access Unlocked Topic	1. Click unlocked 'Arrays' node. 2. Select 'Story Mode'.	Node ID: arrays	Redirects to /play/arrays. Level map loads.	Redirected to /play/arrays. Intro component loaded.	**Pass**
TC_006	**Story Mode**	Render Intro Stage	1. Load intro stage. 2. Verify title, text, and visual.	Topic: Arrays	Text reads "What is an Array?" with locker visual.	Rendered correctly with Framer Motion fade-in.	**Pass**

Gantt Chart



Future Scope

- Expand platform to cover advanced algorithms and competitive programming topics.
- Integrate traditional coding exercises alongside interactive visualizations for deeper understanding.
- Develop collaborative puzzles and guided challenges to encourage peer learning and problem-solving.